

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	MKU201Team76ab4df35a238dc999eb5e6de
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Project: Automated Standard Laptop procurement using ServiceNow Flow Designer

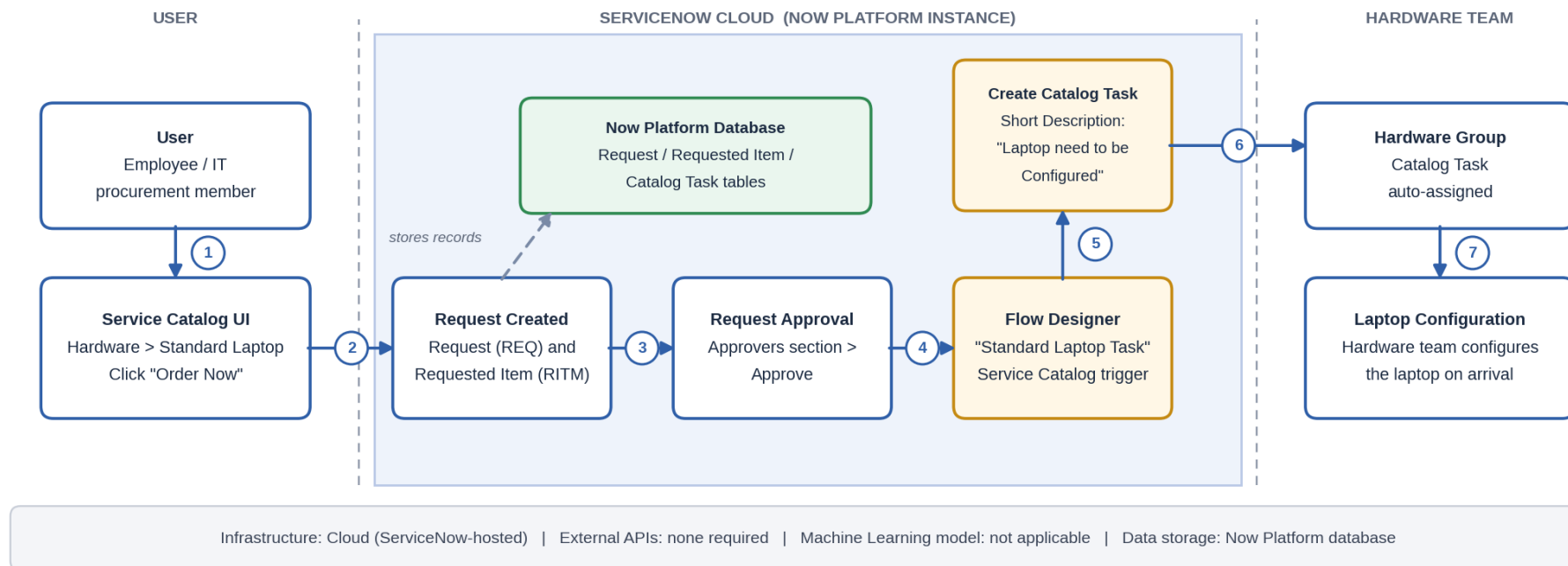


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employees / IT procurement users order the Standard Laptop through the ServiceNow Service Catalog (Hardware > Standard Laptop > Order Now) and check the Order Status.	ServiceNow Service Catalog (Now Platform web UI)
2.	Application Logic-1	"Standard Laptop Task" flow created in the Global application, run as System User, with Service Catalog as the trigger.	ServiceNow Flow Designer
3.	Application Logic-2	Create Catalog Task action mapped to the Requested Item record; Short Description and Description set to "Laptop need to be Configured"; Approval set to Approved.	Flow Designer action: Create Catalog Task
4.	Application Logic-3	Request approved by the approver, and the catalog task is automatically assigned to the Hardware group for laptop configuration.	ServiceNow Approvals and Assignment Group (Hardware)
5.	Database	Stores the Request, Requested Item and Catalog Task records generated by each laptop order.	ServiceNow Now Platform database
6.	Cloud Database	Database service hosted with the ServiceNow cloud instance.	ServiceNow Cloud Instance
7.	File Storage	No separate file storage required; record data is kept within the platform.	ServiceNow Now Platform (built-in storage)
8.	External API-1	Not required; the whole procurement workflow runs inside ServiceNow.	Not applicable
9.	External API-2	Not required.	Not applicable
10.	Machine Learning Model	Not used in this project.	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on Cloud (ServiceNow instance accessed through a web browser) Local Server Configuration: Not required Cloud Server Configuration: ServiceNow-hosted instance	Cloud (ServiceNow Now Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework required; the solution is built with the low-code Flow Designer.	ServiceNow Flow Designer (low-code)
2.	Security Implementations	Flow runs as System User; catalog task restricted to the Hardware group; request needs approval by the approver; role-based access on records.	ServiceNow roles and access controls, approvals

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Cloud platform with separate presentation (catalog), logic (flow) and data layers; the same flow pattern can be reused for other catalog items.	ServiceNow cloud platform, Flow Designer
4.	Availability	Runs on the ServiceNow-hosted cloud instance; once activated the flow is available for every new Standard Laptop request.	ServiceNow cloud instance
5.	Performance	Catalog tasks are created automatically by the flow without manual steps, reducing delays and user wait time.	Event-driven Flow Designer automation

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>